Quantifying Dataset Leakage in Fake News Detection

A Case Study on the ISOT Dataset

Venkateswarlu Nagineni*, Raj Purohith Arjun*, Hareen Sai Vattikuti*
*Department of Computer Science and Engineering
Texas A&M University, College Station, TX, USA
{venkates2002, raj2001, hareen}@tamu.edu

***Abstract*—This paper studies fake news detection as a binary text classification problem using the ISOT Fake News Dataset. The goal is to classify news articles as fake or real based on their textual content.**

We develop a machine learning pipeline that includes text preprocessing, TF-IDF feature extraction, and linguistic features such as readability and sentiment. Multiple models are evaluated, including Logistic Regression, Support Vector Machines, Random Forests, and LightGBM, using cross-validation and statistical evaluation.

The main contribution of this work is a systematic analysis of dataset leakage. We show that commonly reported near-perfect accuracy on the ISOT dataset is partly driven by dataset artifacts, including subject labels and source identifiers such as "Reuters", which can independently achieve accuracy above 99%.

However, after removing these artifacts through a controlled decontamination process, model performance remains high, with linear models achieving F1-scores above 0.98. This indicates that predictive signals extend beyond simple source-based leakage and are embedded in the textual content itself.

These findings highlight both the limitations of current benchmark datasets and the importance of careful dataset auditing and evaluation in fake news detection.

***Index Terms*—Fake news detection, binary text classification, TF-IDF, support vector machine, random forest, ISOT dataset, dataset leakage, decontamination, NLP, misinformation.**

I. Introduction

The rapid growth of online information has significantly increased the spread of misinformation across digital platforms. False or misleading news can reach large audiences quickly, often before it can be verified. Prior work has shown that false information spreads faster and more widely than true information, raising concerns about its societal impact [1].

Automated fake news detection has therefore become an important area of research. In most approaches, the task is formulated as a supervised binary classification problem, where a model predicts whether a news article is fake or real based on its textual content. A wide range of machine learning techniques have been applied, including traditional models such as Logistic Regression and Support Vector Machines, as well as deep learning approaches [2]–[4].

Despite this progress, many studies report very high performance on benchmark datasets, often achieving accuracy and F1-scores above 95%. While these results appear promising, recent work suggests that such performance may be influenced by dataset-specific artefacts rather than genuine language understanding. This phenomenon is related to shortcut learning, where models exploit superficial patterns instead of learning meaningful representations [5]. Similar issues have been observed in other NLP tasks, where annotation artefacts and dataset biases lead to inflated performance [6], [7].

In this work, we examine these issues using the ISOT Fake News Dataset, a widely used benchmark for fake news detection. Although the dataset is commonly used for evaluation, it contains several structural properties that may introduce unintended biases. In particular, the subject field is strongly correlated with the class label, and source-specific markers such as Reuters identifiers are highly imbalanced between fake and real articles. These characteristics create the possibility that models may rely on shortcuts rather than meaningful textual features.

To investigate this, we perform a detailed dataset audit to identify and quantify potential sources of leakage. Our analysis shows that a classifier using only the field achieves perfect classification accuracy, and that a simple binary feature indicating the presence of the word "Reuters" achieves over 99% accuracy. These findings demonstrate that naive evaluation on this dataset can be misleading.

To address this issue, we propose a controlled experimental framework based on three feature representations. The first uses raw text features and reflects a naive baseline. The second removes source-related artefacts through a decontamination process, providing a more reliable estimate of performance. The third uses only linguistic style features, such as readability and sentiment, to evaluate whether stylistic differences alone are sufficient for classification.

This framework enables direct comparison between leakage-driven performance and genuine linguistic signals. Our results show that while dataset artefacts contribute to performance, their removal results in only a minimal decrease in accuracy. This suggests that predictive signals extend beyond simple source identifiers and are embedded in the textual content itself.

The main contributions of this work are as follows:

- A systematic audit of the ISOT dataset, identifying multiple sources of leakage including subject-based separation and source-specific patterns.
- A controlled experimental framework that isolates leakage-driven performance from genuine linguistic learning.

All authors contributed equally to this work.

- An empirical analysis showing that model performance remains high even after decontamination, highlighting the presence of deeper predictive signals.

These findings highlight the importance of careful dataset auditing and evaluation in fake news detection, and suggest that high performance on benchmark datasets should be interpreted with caution.

II. Dataset Description and Structural Audit

A. ISOT Fake News Dataset

This study uses the ISOT Fake News Dataset [8], which consists of two CSV files: `Fake.csv` and `True.csv`. Each file contains four fields: `title`, `text`, `subject`, and `date`.

After merging and assigning labels, the dataset contains 44,898 articles, with 23,481 fake and 21,417 real samples (imbalance ratio ≈ 1.09). No missing values are present in the primary text fields.

Duplicate analysis reveals 6,252 duplicate texts and 6,169 duplicate titles. Additionally, 6,852 near-duplicate articles are identified based on shared prefixes. These redundancies suggest reduced dataset diversity and may inflate model performance.

B. Structural Audit and Leakage Analysis

A detailed audit is conducted to identify dataset artefacts that may bias model performance. Similar issues of dataset artefacts and shortcut learning have been observed in prior NLP studies [5], [7].

1) Subject-Based Leakage: The subject field exhibits complete separation between fake and real articles. Categories such as `News`, `politics`, and `left-news` appear exclusively in fake articles, while `worldnews` and `politicsNews` appear only in real articles.

A classifier trained using only the subject field achieves 100% accuracy, confirming that this feature acts as a direct proxy for the label.

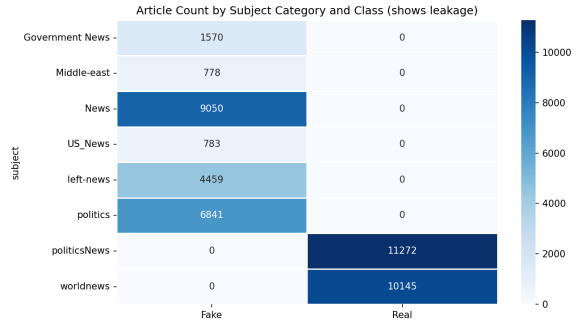

Fig. 1: Distribution of subject categories across fake and real news. The absence of overlap demonstrates strong dataset leakage.

2) Source-Based Leakage (Reuters Fingerprint): A strong source-based pattern is observed. Approximately 99.8% of real articles contain the word "Reuters", compared to only 1.4% of fake articles.

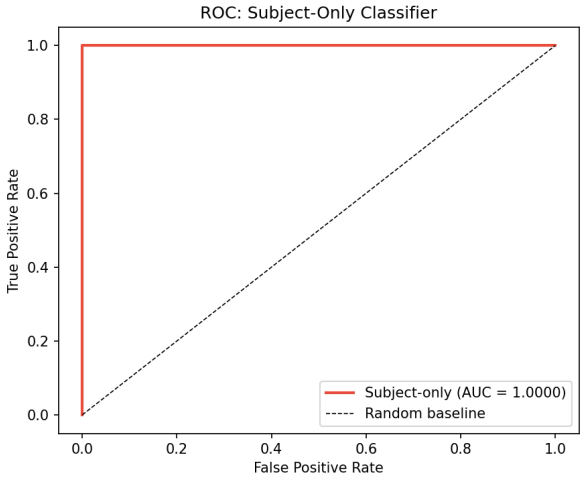

Fig. 2: ROC curve for a classifier using only the subject feature. The perfect AUC confirms complete label leakage.

A simple binary feature indicating the presence of "Reuters" achieves an accuracy of 0.992, demonstrating that models can rely on source identifiers rather than learning meaningful linguistic differences.

3) Temporal Bias: Temporal analysis reveals inconsistencies in the dataset. A total of 33,030 entries contain unparseable or missing dates. Among valid entries, fake articles span 2015–2017, while real articles lack consistent temporal coverage.

This imbalance makes temporal features unreliable and potentially misleading.

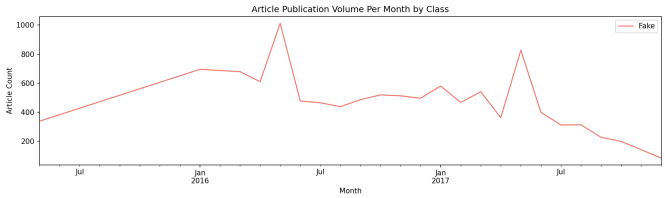

Fig. 3: Monthly article publication volume for fake news. The irregular spikes indicate temporal imbalance in the dataset.

4) Vocabulary and Content Bias: Vocabulary analysis shows clear differences between fake and real articles. Fake news frequently contains tokens such as "https", "21wire", and "flickr", while real news is dominated by terms such as "Reuters", "minister", and "official".

These patterns further indicate that models may exploit dataset-specific artefacts.

C. Summary of Dataset Limitations

Table I summarises the key limitations and their implications.

III. Methodology

A. Research Pipeline Overview

The overall methodology follows a structured pipeline consisting of four main stages: dataset auditing, feature engineering,

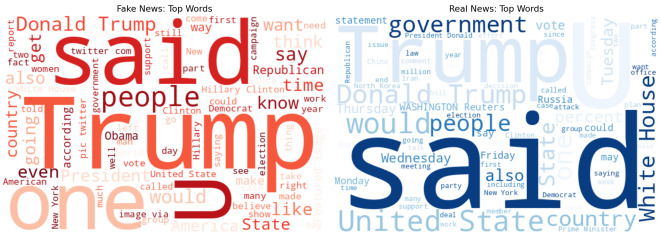

Fig. 4: Word clouds for fake and real news articles. Real news is dominated by structured reporting terms, while fake news contains more informal and promotional language.

TABLE I: Summary of Dataset Limitations

Limitation	Evidence	Impact
Subject leakage	Perfect separation; 100% accuracy	Acts as label proxy; must be excluded
Reuters fingerprint	99.8% real vs 1.4% fake	Models learn source instead of content
Temporal bias	33,030 missing/invalid dates	Date features unreliable
Duplicate content	6,252 duplicate texts; 6,169 titles	Reduces diversity; inflates performance
Limited domain	Primarily US political news	Poor generalisation

model training, and evaluation. Similar structured pipelines are commonly used in fake news detection systems [2].

The analysis begins with a detailed dataset audit to identify potential sources of bias and leakage. Based on this audit, features are constructed under controlled settings to isolate different sources of predictive signal. Multiple machine learning models are then trained using a consistent cross-validation framework.

Finally, models are evaluated using standard performance metrics and statistical tests. This design ensures that performance differences can be attributed to meaningful factors rather than dataset artefacts.

B. Pre-Registered Hypotheses

To avoid post-hoc interpretation, all hypotheses were defined prior to model training. Pre-registration is recommended to reduce bias in empirical machine learning studies [9]. These hypotheses guide both the statistical analysis and the experimental design.

Table II summarises the hypotheses along with the corresponding statistical tests and rejection criteria. .

C. Text Preprocessing

All text data is preprocessed using standard natural language processing techniques commonly used in text classification [3]. Articles are converted to lowercase, tokenised, and stripped of punctuation. Stopwords are removed, and both stemming (Porter Stemmer) and lemmatisation (WordNet) are applied.

A key component of preprocessing is the removal of dataset-specific artefacts identified during the audit phase. In particular,

TABLE II: Pre-Registered Research Hypotheses

ID	Hypothesis	Statistical Test	Rejection Criterion
H1	Linguistic patterns differ significantly between fake and real articles	Mann-Whitney U; Kolmogorov-Smirnov on each continuous feature	$p < 0.05$ with Bonferroni correction
H2	Metadata features carry independent predictive signal beyond text alone	Ablation study + McNemar's test on paired predictions	Significant improvement ($p < 0.05$) when metadata is added
H3	Logistic Regression is statistically competitive with RF and SVM after decontamination	McNemar's test; 95% CI for performance differences	No significant difference ($p > 0.05$) between LR and best complex model

source-identifying patterns such as "Reuters" markers and publication signatures are removed to prevent models from relying on non-linguistic signals.

This decontamination step ensures that subsequent models focus on textual content rather than source-based shortcuts.

D. Feature Engineering—Three Experimental Tracks

To isolate the effect of dataset artefacts, all experiments are conducted using three distinct feature representations (Table III). This design enables direct comparison between leakage-driven performance and genuine linguistic signals.

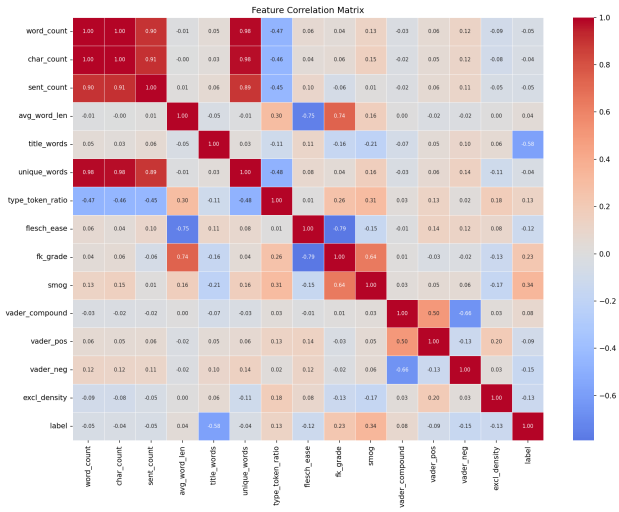

Fig. 5: Feature correlation matrix for engineered features. Most features show weak correlation with the target label, indicating reliance on combined signals.

Track C incorporates 14 linguistically motivated features, including readability scores, sentiment measures, lexical diversity, and punctuation-based features. These features are designed to capture stylistic differences between fake and real news articles.

TABLE III: Three Experimental Feature Tracks

Track	Features	Purpose	Observed F1
A	Raw TF-IDF (50k features; unigrams + bigrams)	Baseline with leakage	0.9916
B	Decontaminated TF-IDF (title + body)	Leakage removed; reliable estimate	0.989–0.995
C	Linguistic features (14 features)	Style-based classification only	0.88–0.93

This three-track formulation is central to the study, as it allows performance differences to be attributed to specific sources of signal.

E. Feature Representation

For text-based features (Tracks A and B), TF-IDF vectorisation is applied using unigram and bigram features, a widely used approach in text classification [2]. A maximum vocabulary size of 50,000 is used, and sublinear term frequency scaling reduces the influence of highly frequent terms.

Due to the high dimensionality of TF-IDF features, dimensionality reduction is applied using Truncated Singular Value Decomposition (SVD), also known as Latent Semantic Analysis (LSA). The feature space is reduced to 300 components.

This representation retains approximately 15.9% of the variance in the data, while significantly reducing computational complexity.

For Track C, features are computed directly from linguistic properties and do not require vectorisation.

F. Model Training

Five model families are evaluated across all three feature tracks, resulting in a 5×3 experimental design. All models are trained using 5-fold stratified cross-validation, with performance reported as mean and standard deviation. A separate 20% hold-out test set is reserved for final evaluation, statistical comparison, and error analysis.

1) Logistic Regression: Logistic Regression is used as the primary interpretable baseline. L2 regularisation is applied due to its stability in high-dimensional feature spaces. L1 regularisation is additionally evaluated for feature selection, as it encourages sparse solutions by driving irrelevant coefficients to zero.

2) Naive Bayes: Multinomial Naive Bayes is evaluated as a probabilistic baseline. Despite its independence assumption, it often performs well on text classification tasks. Complement Naive Bayes is also tested, as it can improve robustness by modelling the complement class distribution.

3) Support Vector Machine: A linear Support Vector Machine is used due to its effectiveness in high-dimensional sparse text data. The regularisation parameter is tuned via grid search, with the optimal value found to be $C = 10$. Probability calibration is applied to enable ROC-AUC evaluation.

4) Tree-Based Models: Random Forest, XGBoost, and LightGBM are used to capture potential non-linear relationships. These models are primarily applied to the LSA-reduced TF-IDF representation and the linguistic feature set (Track C), where interactions between features may be informative.

LightGBM is configured with 300 estimators and a learning rate of 0.05, providing an efficient implementation for structured feature spaces.

IV. Evaluation Framework

Model performance is evaluated using standard classification metrics, including Accuracy, Precision, Recall [10], [11], F1-score, and ROC-AUC. Given the balanced class distribution, F1-score is used as the primary evaluation metric, as it provides a balanced measure of precision and recall.

All results are reported using 5-fold stratified cross-validation [12]. For each model, the mean and standard deviation across folds are computed to ensure robust performance estimates.

A. Statistical Testing

To assess whether observed differences between models are statistically significant, McNemar's test is applied to paired predictions on the test set. This test evaluates whether two models differ significantly in their classification errors.

For feature-level analysis (H1), non-parametric statistical tests are used. The Mann–Whitney U test and Kolmogorov–Smirnov test are applied to compare feature distributions between fake and real articles. A Bonferroni-corrected significance threshold is used to account for multiple comparisons.

B. Hypothesis Evaluation

The experimental design directly supports the evaluation of the pre-registered hypotheses:

- **H1 (Linguistic Differences):** Tested using statistical comparisons of feature distributions, including readability, sentiment, and lexical diversity measures.
- **H2 (Metadata Contribution):** Evaluated through ablation analysis by comparing performance across feature tracks, particularly between Track B (decontaminated text) and Track C (linguistic features only).
- **H3 (Model Comparison):** Assessed using McNemar's test and confidence intervals to compare Logistic Regression with more complex models such as SVM and Random Forest.

C. Error Analysis

In addition to quantitative evaluation, qualitative error analysis is conducted by examining misclassified examples. The Logistic Regression model produces 39 false positives and 62 false negatives on the test set.

False positives typically correspond to articles with informal or sensational language, while false negatives include structured content that resembles legitimate reporting. These cases highlight the limitations of purely text-based models.

V. Results

A. Leakage Quantification

To quantify the impact of dataset artefacts, we compare model performance between Track A (leaky baseline) and Track B (decontaminated text). Table IV shows the F1-scores and corresponding performance differences.

TABLE IV: Leakage Inflation: Track A vs. Track B (F1-score)

Model	Track A	Track B	Difference
LR-L2	0.9916	0.9891	-0.0025
Linear SVM	0.9916*	0.9953	+0.0037
Random Forest	0.985*	0.9719	-0.0131

*Track A approximated where not directly evaluated

The results show that removing source-based artefacts results in only a marginal change in performance. For Logistic Regression, the F1-score decreases by only 0.0025, while SVM achieves slightly higher performance on decontaminated data.

These findings align with prior work on dataset leakage and shortcut learning, which shows that machine learning models often exploit spurious correlations when available rather than learning robust semantic representations [5], [9]. Similar behaviour has been observed in NLP benchmarks, where annotation artefacts can lead to inflated performance without genuine task understanding

The minimal performance drop observed after decontamination suggests that, while leakage contributes to performance, the models are also capturing deeper lexical and contextual signals. This supports the argument that high accuracy on fake news detection tasks may arise from a combination of artefact exploitation and genuine linguistic pattern recognition.

B. Master Results

Linear models (Logistic Regression and SVM) achieve the highest performance on decontaminated data. Tree-based models perform competitively when using reduced-dimensional or structured features, but show larger overfitting gaps.

The strong performance of linear models is consistent with prior findings in text classification, where high-dimensional sparse representations such as TF-IDF often lead to linearly separable decision boundaries [13]. This explains why Logistic Regression and SVM achieve near-optimal performance without requiring complex architectures.

In contrast, tree-based models show relatively lower performance and higher variance, particularly on structured feature sets. This behaviour has been previously attributed to the difficulty of modelling sparse high-dimensional text data using tree-based methods [14]. These results reinforce the suitability of linear models as strong baselines in fake news detection tasks

Table V presents cross-validated performance across all model–track combinations.

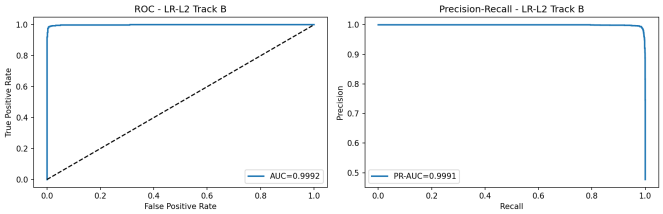

Fig. 6: ROC and precision-recall curves for Logistic Regression on Track B. The model achieves near-perfect performance.

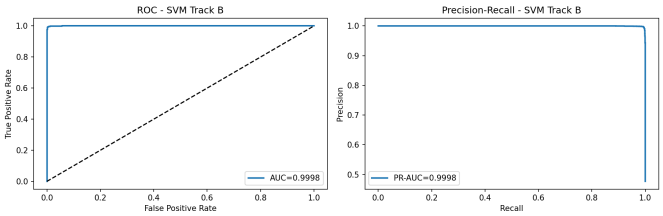

Fig. 7: ROC and precision-recall curves for SVM on Track B. Performance is comparable to Logistic Regression.

C. Hypothesis Resolution

Table VI summarises the evidence and outcomes for the pre-registered hypotheses.

H1 is supported, as multiple linguistic features show statistically significant differences between fake and real articles. H2 is also supported, as linguistic features alone are insufficient to match text-based performance.

H3 is rejected, as Support Vector Machines significantly outperform Logistic Regression, indicating that linear models are not always equivalent in performance.

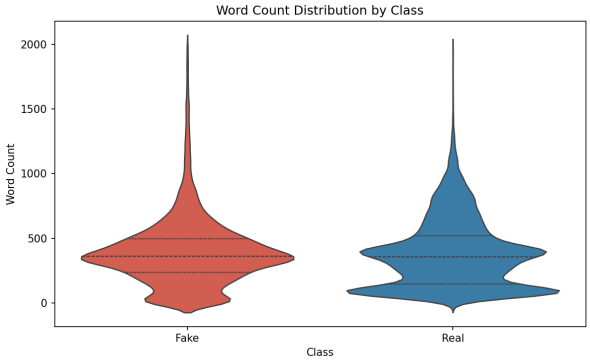

Fig. 8: Distribution of article word counts for fake and real news. Real articles tend to have slightly higher median length, while fake articles show greater variability.

VI. Interpretability and Analysis

Model interpretability is critical in fake news detection, as it enables verification of whether models rely on meaningful linguistic signals or spurious dataset artefacts. Prior work has emphasised that high-performing models may still lack

TABLE V: Cross-Validated Model Performance

Model	Track	Acc.	Prec.	Recall	F1	ROC-AUC	Gap
LR-L2	A	0.9920	0.9903	0.9929	0.9916	0.9993	LOW
LR-L2	B	0.9896	0.9878	0.9904	0.9891	0.9991	LOW
LR-L1	B	0.9888	0.9855	0.9911	0.9883	0.9992	LOW
MNB	B	0.9594	0.9571	0.9579	0.9575	0.9910	LOW
SVM	B	0.9955	0.9956	0.9950	0.9953	0.9997	LOW
RF	C	0.9228	0.9119	0.9279	0.9198	0.9755	HIGH
XGBoost	C	0.9241	0.9051	0.9393	0.9219	0.9774	MED
LightGBM	B-LSA	0.9874	0.9885	0.9851	0.9868	0.9989	MED

TABLE VI: Hypothesis Resolution

ID	Evidence	Test	Verdict
H1	Significant differences in readability, sentiment, and lexical features	Mann–Whitney U + Kolmogorov–Smirnov (KS)	Supported
H2	Large performance gap between text-based (0.99) and style-only models (0.88–0.93)	Ablation analysis	Supported
H3	SVM significantly outperforms Logistic Regression ($p < 0.001$)	McNemar's test	Rejected

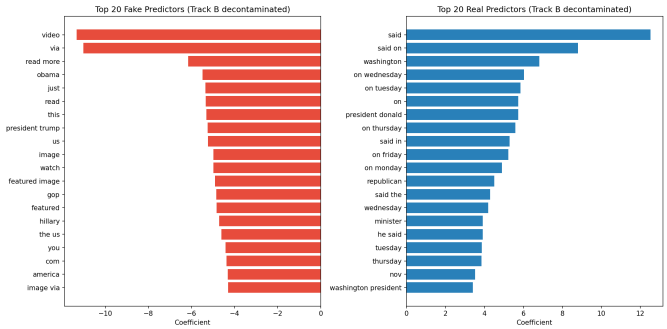

Fig. 9: Top predictive features from Logistic Regression. Real news is associated with structured reporting terms, while fake news includes informal and promotional language.

transparency and robustness, particularly when trained on biased datasets [15]

A. Logistic Regression Coefficient Analysis

Logistic Regression provides a directly interpretable model, where each coefficient represents the change in log-odds associated with a feature. Positive coefficients indicate features associated with real news, while negative coefficients indicate features associated with fake news.

After decontamination, the key question is whether the most important features still reflect source-specific artefacts or capture meaningful linguistic patterns. Analysis of the learned coefficients shows that real news is characterised by formal reporting language (e.g., "said", "minister", "official"), while fake news is associated with more informal and sensational expressions (e.g., "video", "breaking", "watch").

The observed coefficients suggest that the model captures stylistic and structural differences consistent with journalistic writing conventions. Real news tends to follow formal reporting patterns, while fake news exhibits markers of sensationalism and informal discourse. This aligns with prior linguistic studies showing that misinformation often leverages emotional language, exaggeration, and reduced lexical diversity to increase engagement [16], [17].

B. SHAP Analysis for Tree-Based Models

To analyse non-linear models, SHAP (SHapley Additive exPlanations) values are computed for the Random Forest model trained on Track C (linguistic features) [18]. A subset of 500 samples is used for computational efficiency.

SHAP summary plots indicate that features such as readability scores, sentiment polarity, and lexical diversity have the largest influence on predictions. For example, higher sentiment polarity and increased punctuation density tend to push predictions towards the fake class, while higher readability complexity and vocabulary richness favour the real class.

These results confirm that non-linear models capture interactions between linguistic features, although their overall predictive power remains lower than text-based representations.

C. Linguistic Feature Importance

Table VII summarises the linguistic features used in Track C and their expected directional influence. These features are grounded in prior work on misinformation and writing style.

Overall, these analyses provide insight into the types of patterns learned by different models and support the interpretation of the quantitative results.

VII. Discussion

A. Impact of Dataset Leakage

The dataset audit confirms the presence of strong leakage in the ISOT dataset [8], particularly through subject labels and source identifiers such as "Reuters". These features alone achieve near-perfect classification accuracy, demonstrating that naive models can rely on shortcuts rather than meaningful language understanding.

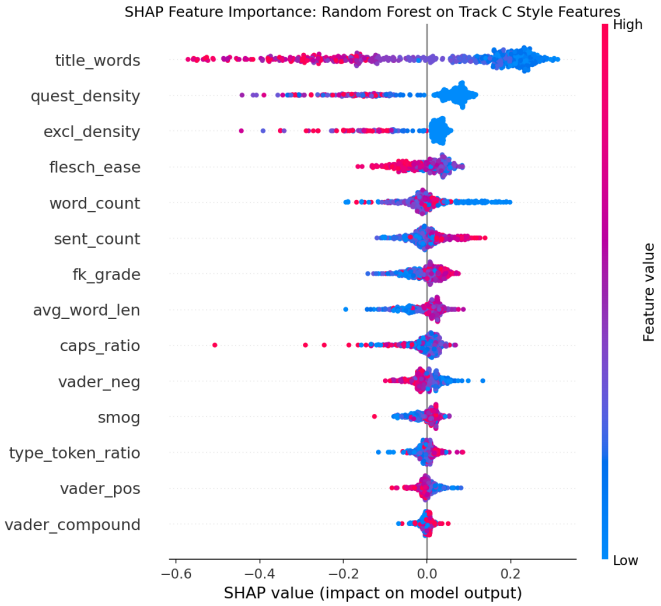

Fig. 10: SHAP summary plot for Random Forest on Track C features. Feature importance and direction indicate the influence of writing style characteristics.

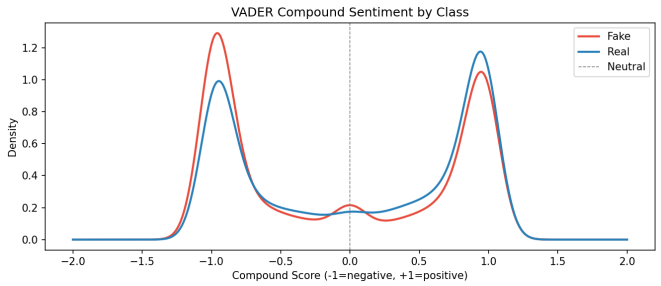

Fig. 11: Distribution of VADER compound sentiment scores. Fake news exhibits more extreme sentiment values compared to real news.

However, the experimental results show that removing these artefacts leads to only a minimal reduction in performance. The difference between Track A (leaky) and Track B (decontaminated) is small, with Logistic Regression dropping from an F1-score of 0.9916 to 0.9891. This indicates that while leakage exists, it is not the sole factor driving high performance.

B. Key Findings

Several important findings emerge from this analysis.

First, dataset artefacts such as subject labels and source identifiers can produce near-perfect classification performance. This confirms that evaluation without proper dataset auditing can be misleading.

Second, removing these artefacts results in only a small decrease in performance. This suggests that predictive signals extend beyond simple source fingerprints and are embedded in the text itself.

TABLE VII: Track C Linguistic Features and Directional Hypotheses

Feature	Linguistic Basis	Expected Direction
Flesch Reading Ease	Simpler language increases accessibility and spread	Fake: higher ease score
Type-Token Ratio (TTR)	Rich vocabulary reflects formal reporting	Real: higher TTR
VADER Compound Sentiment	Emotional amplification in fake news	Fake: more extreme polarity
Exclamation Density	Marker of sensationalism	Fake: higher density
SMOG Index	Complexity indicates domain expertise	Real: higher SMOG
Capitalisation Ratio	Visual emphasis in informal writing	Fake: higher caps ratio
Question Mark Density	Rhetorical framing in persuasive content	Fake: higher density
Avg. Sentence Length	Structured writing in journalism	Real: longer sentences

Third, linguistic features capture meaningful stylistic differences between fake and real news. However, models trained only on these features achieve lower performance (F1 0.88–0.93), indicating that writing style alone is insufficient for accurate classification.

Finally, linear models such as Logistic Regression and Support Vector Machines perform as well as or better than more complex models. This suggests that the classification boundary is largely linear in TF-IDF feature space, and that increased model complexity does not necessarily yield better performance.

C. Why Does Performance Remain High

The sustained high performance after decontamination can be explained by systematic differences in both content and structure between fake and real news.

Real news articles tend to follow formal journalistic conventions, using consistent reporting language such as “said”, “official”, and “minister”. In contrast, fake news articles often include more informal or sensational expressions, multimedia references, and non-standard formatting.

Additionally, topic distribution plays a role. Real news frequently focuses on international politics and policy, while fake news often emphasises opinion-driven or domestic narratives. These differences create patterns that models can learn even in the absence of explicit source identifiers.

D. Model Behaviour and Feature Learning

The results indicate that models rely heavily on lexical and contextual patterns rather than purely stylistic features. TF-IDF [2], [19] representations capture both vocabulary usage and local context, enabling models to distinguish between classes effectively.

Tree-based models, while capable of capturing non-linear interactions, do not consistently outperform linear models and

show larger overfitting gaps. This suggests that the primary signals in the dataset are linearly separable and that simpler models are sufficient for this task.

E. Implications for Evaluation

These findings have important implications for the evaluation of fake news detection systems. High performance on benchmark datasets does not necessarily indicate robust real-world performance, particularly when datasets contain structural artefacts or domain biases.

This study highlights the importance of dataset auditing, controlled experimental design, and careful interpretation of results. Without these steps, models may appear highly accurate while relying on dataset-specific shortcuts that do not generalise.

VIII. Conclusion

This study investigated fake news detection using the ISOT dataset, with a focus on understanding the impact of dataset artefacts on model performance.

A detailed dataset audit revealed strong sources of leakage, including subject-based separation and source identifiers such as "Reuters", which can independently achieve near-perfect classification accuracy. To address this, a controlled experimental framework was introduced using three feature tracks, enabling direct comparison between leakage-driven performance and genuine linguistic signals.

The results show that while dataset artefacts contribute to performance, their removal leads to only a minimal decrease in accuracy. Models such as Logistic Regression and Support Vector Machines maintain F1-scores above 0.98 even after decontamination, indicating that predictive signals extend beyond simple source identifiers. At the same time, linguistic features alone achieve lower performance, suggesting that writing style is informative but not sufficient for accurate classification.

These findings highlight the importance of careful dataset auditing and controlled evaluation when developing fake news detection systems. High performance on benchmark datasets should be interpreted with caution, particularly when structural biases are present.

A. Limitations and Future Work

This study is limited to a single dataset focused on US political news, which may restrict the generalisability of the findings. Additionally, the analysis considers only textual features and does not incorporate contextual information such as user behaviour or network propagation.

Future work could extend this analysis to more diverse datasets and domains, incorporate contextual and temporal features, and develop benchmark datasets with reduced bias and improved annotation quality.

References

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